problems with the pilot-wave theory. Pauli wrote back next, throwing those same problems at Bohm. But Bohm managed to handle them with style and aplomb through his brilliant insight that measurement devices themselves must be incorporated in his quantum descriptions. Pauli, after a lengthy and heated exchange of letters over the next few months, finally conceded that Bohm's theory was consistent, though he still maintained that, since there was no way to test it against "normal" quantum physics, it remained "a check that cannot be cashed." Ultimately, Pauli thought that Bohm's ideas were simply "artificial metaphysics."

Niels Bohr himself never wrote back to Bohm. But Bohm did receive a report from his friend, Art Wightman, who was visiting Bohr's institute at the time. According to Wightman, Bohr thought that Bohm's theory was "very foolish," and didn't say much else. Von Neumann, meanwhile, was less dismissive; he thought Bohm's ideas were "consistent," and even "very elegant," but suspected that Bohm would have difficulties extending his theory to encompass the quantum phenomenon of spin—a suspicion that ultimately proved to be wrong.

Von Neumann's suspicion was likely born of his own "impossibility" proof demonstrating the necessity of the Copenhagen interpretation. Bohm knew that his own theory showed that something was wrong with that proof—or at least that it was a less powerful proof than commonly supposed by other physicists. Bohm discussed how his theory evaded von Neumann's proof toward the end of his second paper laying out his pilot-wave theory. But his analysis of von Neumann's proof was somewhat unclear at best and simply incorrect at worst. And without a clear and pithy explanation of what had gone wrong in von Neumann's proof, many physicists assumed the flaw was with Bohm's theory instead—that it simply couldn't be correct, because von Neumann had shown such theories were impossible.

There were a few physicists who did come around to supporting Bohm's view: most notably, Louis de Broglie, who took up his old interpretation and started a priority dispute with Bohm over the work. Bohm resisted acknowledging de Broglie's contributions at first. "If one man finds a diamond and then throws it away because he falsely concludes it is a valueless stone, and if this stone is later found by another man who recognizes its true value, would you not say that the stone belongs to the second man?" Nonetheless, the dispute was short-lived and resolved